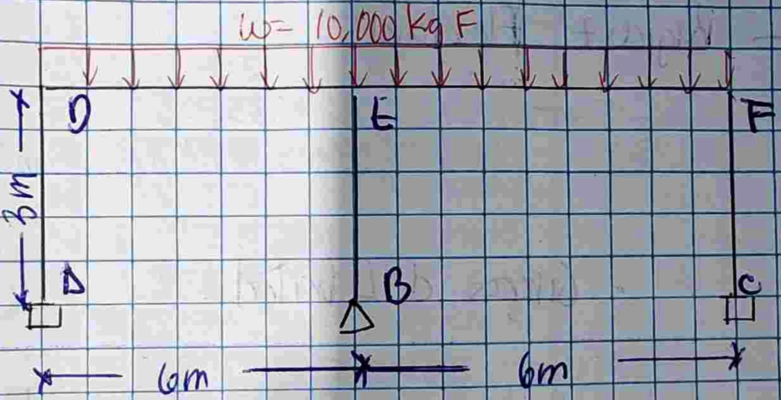
Marlon Ivan Carreto Rivera
201230088.

Serie I.

Construya su matriz de rigidez y Reduzcala



- Datos:
- → $E = 250,000\ kgF/cm^2$
- → Sección para vigas y columnas:
  - b = 50 cm
  - a = 50 cm
- → Longitud L = 6m.
- → W = 10 000 kgF
- → Altura de Columna = 300 cm
- → Largo entre vigas = 600 cm

- Propiedades de la Sección:

$$I = \frac{bd^3}{12} = \frac{50(50)^3}{12}$$

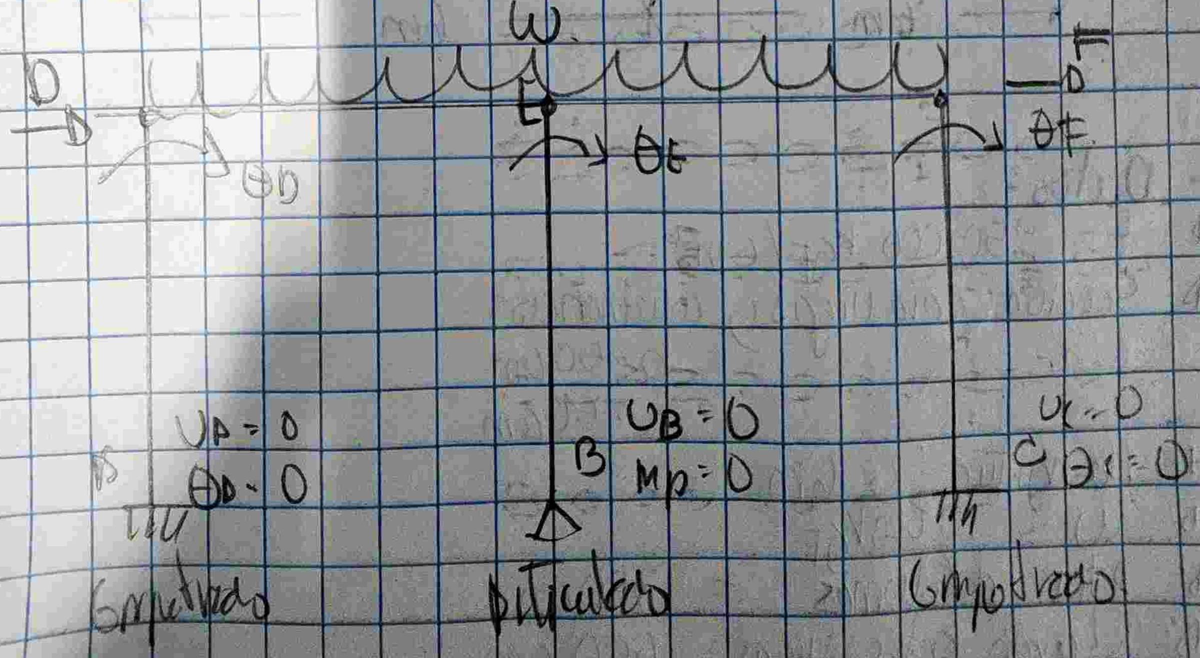
Momento de Inercia

$$I = 520\,833.33 \ cm^4$$

- Rigidez Flexional:

$$EI = 250000\,(520,833.33)$$
$$EI = 130\,20\,83333 \times 10^{11} \ kgF \cdot cm^2$$

Grados de Libertad.



Matriz Global 12X12.

	A1	A2	B1	B2	C1	C2	D1	D2	E1	E2	F1	F2
A1	$4h^2$	$6h$	0	0	0	0	$2h^2$	$6h$	0	0	0	0
A2	$-6h$	-12	0	0	0	0	$-6h$	-12	0	0	0	0
B1	0	0	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$	0	0
B2	0	0	$-6h$	12	0	0	0	0	$-6h$	-12	0	0
C1	0	0	0	0	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$
C2	0	0	0	0	$-6h$	12	0	0	0	0	$-6h$	-12
D1	$2h^2$	$-6h$	0	0	0	0	$6h^2$	$6h$	h^2	0	0	0
D2	$6h$	-12	0	0	0	0	$6h$	12	0	0	0	0
E1	0	0	$2h^2$	$-6h$	0	0	h^2	0	$8h^2$	$6h$	h^2	0
E2	0	0	$6h$	-12	0	0	0	0	$6h$	12	0	0
F1	0	0	0	0	$2h^2$	$-6h$	0	0	h^2	0	$6h^2$	$6h$
F2	0	0	0	0	$6h$	-12	0	0	0	0	$6h$	12

* Eliminación de A1.

* Eliminación de A2.

	A_2	B_1	B_2	C_1	C_2	D_1	D_2	E_1	E_2	F_1	F_2
A_2	12	0	0	0	0	$6h$	12	0	0	0	0
B_1	0	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$	0	0
B_2	0	$-6h$	12	0	0	0	0	$-6h$	-12	0	0
C_1	0	0	0	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$
C_2	0	0	0	$-6h$	12	0	0	0	0	$-6h$	-12
D_1	$-6h$	0	0	0	0	$6h^2$	$6h$	h^2	0	0	0
D_2	-12	0	0	0	0	$6h$	12	0	0	0	0
E_1	0	$2h^2$	$-6h$	0	0	h^2	0	$8h^2$	$6h$	h^2	0
E_2	0	$6h$	-12	0	0	0	0	$6h$	12	0	0
F_1	0	0	0	$2h^2$	$-6h$	0	0	h^2	0	$6h^2$	$6h$
F_2	0	0	0	$6h$	-12	0	0	0	0	$6h$	12

* Eliminación de B_2

	B_1	B_2	C_1	C_2	D_1	D_2	E_1	E_2	F_1	F_2
B_1	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$	0	0
B_2	$6h$	12	0	0	0	0	$6h$	12	0	0
C_1	0	0	$4h^2$	$-6h$	0	0	0	0	$2h^2$	$6h$
C_2	0	0	$-6h$	12	0	0	0	0	$-6h$	-12
D_1	0	0	0	0	$6h^2$	$6h$	h^2	0	0	0
D_2	0	0	0	0	$6h$	12	0	0	0	0
E_1	$2h^2$	$-6h$	0	0	h^2	0	$8h^2$	$6h$	h^2	0
E_2	$6h$	-12	0	0	0	0	$6h$	12	0	0
F_1	0	0	$2h^2$	$-6h$	0	0	h^2	0	$6h^2$	$6h$
F_2	0	0	$6h$	-12	0	0	0	0	$6h$	12

* Eliminación de C_1

	B_1	C_1	C_2	D_1	D_2	E_1	E_2	F_1	F_2
B_1	$4h^2$	0	0	0	0	$2h^2$	$6h$	0	0
C_1	0	$4h^2$	$6h$	0	0	0	0	$2h^2$	$6h$
C_2	0	$-6h$	12	0	0	0	0	$6h$	-12
D_1	0	0	0	$6h^2$	$6h$	h^2	0	0	0
D_2	0	0	0	$6h$	12	0	0	0	0
E_1	$2h^2$	0	0	h^2	0	$8h^2$	$6h$	h^2	0
E_2	$6h$	0	0	0	0	$6h$	12	0	0
F_1	0	$2h^2$	$-6h$	0	0	h^2	0	$6h^2$	$6h$
F_2	0	$6h$	-12	0	0	0	0	$6h$	12

* Eliminación de C2.

	B_1	C_2	D_1	D_2	E_1	E_2	F_1	F_2
B_1	$4h^2$	0	0	0	$2h^2$	$6h$	0	0
C_2	0	12	0	0	0	0	$6h$	12
D_1	0	0	$6h^2$	$6h$	h^2	0	0	0
D_2	0	0	$6h$	12	0	0	0	0
E_1	$2h^2$	0	h^2	0	$8h^2$	$6h$	h^2	0
E_2	$6h$	0	0	0	$6h$	12	0	0
F_1	0	$-6h$	0	0	h^2	0	$6h^2$	$6h$
F_2	0	-12	0	0	0	0	h	12

* Matriz Libre 7x7. después d Aplicar Apoyos

	B_1	D_1	D_2	E_1	E_2	F_1	F_2
B_1	$4h^2$	0	0	$2h^2$	$6h$	0	0
D_1	0	$6h^2$	$6h$	h^2	0	0	0
D_2	0	$6h$	12	0	0	0	0
E_1	$2h^2$	h^2	0	$8h^2$	$6h$	h^2	0
E_2	$6h$	0	0	$6h$	12	0	0
F_1	0	0	0	h^2	0	$6h^2$	$6h$
F_2	0	0	0	0	0	$6h$	12.

Reordenamiento Matriz Libre
7x7.

	D_2	E_2	F_2	D_1	E_1	F_1	B_1
D_2	12	0	0	6h	0	0	0
E_2	0	12	0	0	6h	0	6h
F_2	0	0	12	0	0	6h	0
D_1	6h	0	0	$6h^2$	h^2	0	0
E_1	0	6h	0	h^2	$8h^2$	h^2	$2h^2$
F_1	0	0	6h	0	h^2	$6h^2$	0
B_1	0	6h	0	0	$2h^2$	0	$4h^2$

* Unificación Columna

$D_2 + E_2$

	D_2	E_2	F_2	D_1	E_1	F_1	B_1
D_2	12	0	0	$6h$	0	0	0
E_2	0	12	0	0	$6h$	0	$6h$
F_2	0	0	12	0	0	$6h$	0
D_1	$6h$	0	0	$6h^2$	h^2	0	0
E_1	0	$6h$	0	h^2	$8h^2$	h^2	$2h^2$
F_1	0	0	$6h$	0	h^2	$6h^2$	0
B_1	0	$6h$	0	0	$2h^2$	0	$4h^2$

Matriz 7x6

$D_2 + E_2$

	D_2	F_2	D_1	E_1	F_1	B_1
D_2	12	0	$6h$	0	0	0
E_1	12	0	0	$6h$	0	$6h$
F_2	0	12	0	0	$6h$	0
D_1	$6h$	0	$6h^2$	4^2	0	0
E_1	$6h$	0	4^2	$8h^2$	h^2	$2h^2$
F_1	0	$6h$	0	h^2	$6h^2$	0
B_1	$6h$	0	0	$2h^2$	0	$4h^2$

* Unificación

$A_2 + F_2$

	A_2	F_2	D_1	E_1	F_1	B_1
D_2	12	0	$6h$	0	6	6
E_2	12	0	0	$6h$	0	$6h$
F_2	0	12	0	0	$6h$	6
D_1	$6h$	0	$6h^2$	h^2	0	0
E_1	$6h$	0	h^2	$8h^2$	h^2	$2h^2$
F_1	0	$6h$	0	h^2	$6h^2$	0
B_1	$6h$	0	0	$2h^2$	0	$4h^2$

Matriz intermedia

7x5

Columnas compatibles ya unificados

	A	D1	E1	F1	B1
D2	12	6h	0	0	0
E2	12	0	6h	0	6h
F2	12	0	0	6h	0
D1	6h	$6h^2$	h^2	0	0
E1	6h	h^2	$8h^2$	h^2	$2h^2$
F1	6h	0	h^2	$6h^2$	0
B1	6h	0	$2h^2$	0	$4h^2$

Unificación de Filas.

D_2 + E_2

	Δ	D_1	E_1	F_1	B_1
D_2	12	$6n$	0	0	0
E_2	12	0	$6n$	0	$6n$
F_2	12	0	0	$6n$	0
D_1	$6n$	$6n^2$	n^2	0	0
E_1	$6n$	n^2	$8n^2$	n^2	$2n^2$
F_1	$6n$	0	n^2	$6n^2$	0
B_1	$6n$	0	$2n^2$	0	$4n^2$

* Unificación de filas
Δ_2 + F_2

	Δ	D_1	E_1	F_1	B_1
Δ 12	24	$6n$	$6n$	0	$6n$
F_2	12	0	0	$6n$	0
D_1	$6n$	$6n^2$	n^2	0	0
E_1	$6n$	n^2	$8n^2$	n^2	$2n^2$
F_1	$6n$	0	n^2	$6n^2$	0
B_1	$6n$	0	$2n^2$	0	$4n^2$

* Matriz Reducida después de Compatibilidad.

	Δ	D_1	E_1	F_1	B_1
Δ	36	$6n$	$6n$	$6n$	$6n$
D_1	$6n$	$6n^2$	$6n$	0	0
E_1	$6n$	n^2	$8n^2$	n^2	$2n^2$
F_1	$6n$	0	n^2	$6n^2$	0
B_1	$6n$	0	$2n^2$	0	$4n^2$

+ Condensación Matricial por bloques

$$K_c = K_{aa} - K_{ab} K_{bb}^{-1} K_{ba}$$

$$k_c = \frac{EI}{h^3}\left(\frac{297}{20}\right) = 71614.5833 \text{ kgf/cm}$$

+ Condensación S1:
Eliminar el giro B1

	Δ	θ_1	E_1	ξ_1	B_1
Δ	36	6h	6h	6h	6h
θ_1	6h	$6h^2$	h^2	0	0
E_1	6h	h^2	$8h^2$	h^2	$2h^2$
ξ_1	6h	0	h^2	$6h^2$	0
B_1	6h	0	$2h^2$	0	$4h^2$

+ Resultado S1: Matriz 4x4.

	Δ	θ_1	E_1	ξ_1
Δ	27	6h	3h	6h
θ_1	6h	$6h^2$	h^2	0
E_1	3h	h^2	$7h^2$	h^2
ξ_1	6h	0	h^2	$6h^2$

* Condensar D1.

	A	D1	E1	F1
A	27	$6h$	$3h$	$6h$
D1	$6h$	$6h^2$	h^2	0
E1	$3h$	h^2	$7h^2$	h^2
F1	$6h$	0	h^2	$6h^2$

* Resultado S2: Matriz 3X3.
Después de Condensar D1

	A	E1	F1
A	21	$9h$	$6h$
E1	$9h$	$41h^2/6$	h^2
F1	$6h$	h^2	$6h^2$

* Condensación S 3: Eliminar el giro F1

	A	E1	F1
A	21	$9h$	$6h$
E1	$9h$	$41h^2/6$	h^2
F1	$6h$	h^2	$6h^2$

x Resultado S3

Matriz 2x2.

	Δ	E1
Δ	15	h
E1	h	$20h^2/3$

Condensación S4: Eliminar el giro E1

	Δ	E1
Δ	15	h
E1	h	$20h^2/3$

Resultado S4:

Matriz 1x1 después de condensar E1.

	Δ
Δ	297/10.

Masa y Propiedades Dinamicas

$m = \frac{w}{g} = \frac{1000}{981}$ $\qquad k_c = 7161.45833\ kgf/cm$

$m = 1.01936799\ kgf\,s^2/cm$

$m\ddot{u}(t) + k_c u(t) = 0$

$1.01936799\,\ddot{u}(t) + 7161.45833\,u(t) = 0$

$\ddot{u}(t) + 7025.390625\,u(t) = 0$

$\omega_n = \sqrt{\frac{k_c}{m}} = \sqrt{\frac{7161.458\,kgf}{1.01936799}} = 83.8176\ rad/s$

$T_n = 2\pi/\omega_n = 2\pi/83.817 = 0.074\,s.$

$f_n = 1/T = 1/0.074 = 13.34\ Hz.$

R//	$\omega_n = 83.8176\ rad/s$
R//	$f_n = 13.34\ Hz$
R//	$T_n = 0.074963\,s.$